



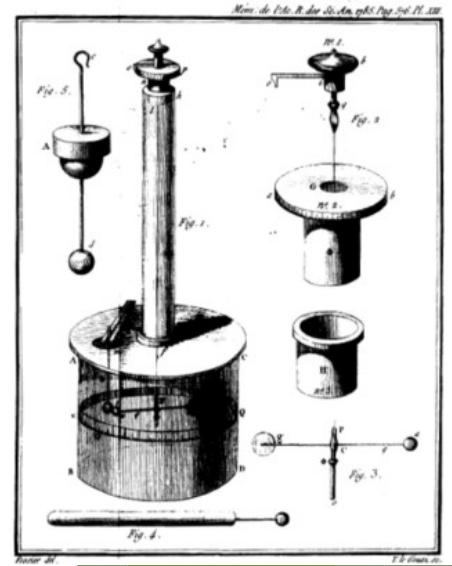
Coulomb's *Law - Electric* *Fields (13-15)*

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Coulomb's Law

- Charles Coulomb 1785
- Torsion Balance(charged rod suspended by string & can be twisted by elec-forces from the charge)
- Macro-scale is big for subatomics
- Electric force proportional to charge (inverse square)

$$F = \frac{kq_1q_2}{r^2}$$



Coulomb's Law_(pt.2)

- Electron is - 1.7×10^{-19} coulombs (quantized)
- Static electricity -> micro/nanocoulombs
- Coulomb's constant is $8.99 \times 10^9 \text{ N m}^2/\text{C}^2$
- Electric force is stronger than gravitational, but most atoms are neutral



Coulomb's constant
($8.99 \times 10^9 \text{ N m}^2/\text{C}^2$)

Charge of both
objects

Electric
force

$$F = \frac{k q_1 q_2}{r^2}$$

distance

A decorative graphic at the bottom right of the slide consisting of several green geometric shapes, including a triangle and a rectangle, arranged in a stepped pattern.

Example from packet

- Rubbing glass rod with cloth takes the glass' electrons and moves them to the cloth (protons stay where they are)
- total charge for the glass & cloth system is 0 C
- Total charge of universe is also 0 C



I thank

you

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